HAOYAN ZHENG

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EDUCATION

B.A.Sc. in Engineering Science (Robotics) + Professional Experience Year (PEY) Sep. 2022 – Jun. 2027
University of Toronto, Toronto, ON

- **Major:** Robotics Engineering | **Minor:** Artificial Intelligence
- **Relevant Courses:** Algorithms & Data Structures (Python, C/C++); Digital Systems & Computer Organization (SystemVerilog, Assembly); Electromagnetism; Microprocessors & Embedded Systems; Unsupervised Machine Learning.
- **Software:** Microsoft Azure Storage Explorer (Blob/Data Lake), ArcGIS, AutoCAD, ModelSim, Quartus, Git, Visual Studio Code, LTspice, MATLAB, Covidence, Microsoft 365.
- **Languages/Frameworks:** Python, Java, C/C++, MATLAB, SystemVerilog, Assembly, ROS.

EXPERIENCE

Pavement Evaluation Engineer May 2025 – Present
Ministry of Transportation, North York, ON

- Consolidated multi-source pavement datasets (distress surveys, LCMS/vision outputs, FWD, GPR, GPS) into Azure Blob Storage; established **4× redundancy** (**2 on-prem physical**, **2 cloud**) with object versioning and WORM retention; standardized container hierarchy, metadata schema, and naming conventions; enforced **least-privilege** RBAC/SAS access; automated **SHA-256** checksums with manifest logging; applied lifecycle rules (hot→cool/archive) with **audit trails** to streamline analysis and governance.
- Built a clustering pipeline (Python, scikit-learn) across ~**60,000** road sections to prioritize **10** representative test sites; improved geographic/structural coverage while reducing expected field mileage and mobilization.
- Developed Power Automate workflows for technician submissions: automated acknowledgements, data validation, and distress index computation; achieved **real-time** responses and reduced manual admin by **8–10 hrs/week**.
- Modeled friction–speed relationships with polynomial least squares; reported $R^2 = 0.92$ with prediction intervals, residual checks, and Q–Q plots; informed **network-level safety thresholds** and policy recommendations.
- Supported computer-vision crack identification using iVision LCMS (LeNet/U-Net); delivered QA/QC dashboards with confusion matrices (**mIoU 0.78**, **F1 0.82**) to guide retraining and ground-truthing.
- Prepared ArcGIS web maps and analytical visuals for program reviews; translated findings into **rehabilitation recommendations** and scheduling inputs for the pavement management system.

Research Analyst Intern May 2024 – May 2025
Michael Garron Hospital, East York, ON

- Contributed to an **AHRQ-funded** Diagnostic Center of Excellence on diagnostic safety and human factors; coordinated literature syntheses across PubMed, Scopus, MEDLINE, and Embase.
- Screened, reviewed, and extracted data from **10,000+** publications using Covidence; authored PRISMA flows and data dictionaries to improve reproducibility and auditability.
- Drafted **REB** protocols and materials (recruitment scripts, consent forms, instruments) for patient interviews; ensured ethics compliance and standardized data capture.
- Designed **WHO Patient Safety Day** engagement activities to raise awareness of diagnostic safety among patients, families, and clinicians.

PROJECTS

Patient Survey Analysis Using Unsupervised Machine Learning *Python, nltk, Gensim, hyperopt, guidedlda*

- Built an LDA topic-modeling pipeline to identify themes from patient free-text survey responses in < **0.05 s/record**; integrated preprocessing (stopword removal, spelling correction, cleaning, POS tagging, lemmatization, tokenization).
- Tuned **10** hyperparameters with TPE (Bayesian optimization) to improve coherence/perplexity; validated topics with quantitative metrics and qualitative review.
- Trained/evaluated on **1,500** survey responses; exported interpretable topic labels and example excerpts for stakeholder consumption.

Autonomous Delivery Robot *ROS, Python*

- Implemented Bayesian localization and line-following for TurtleBot3; completed multi-stop delivery on a closed loop at **0.1 m/s** with reliable waypoint arrival and recovery behaviors.
- Developed simulation for measure–plan–act loops; packaged launch files and configs for repeatable runs and benchmarking.

ACHIEVEMENTS

UTMIST Hackathon — Winner *University of Toronto* Mar. 2023
Top team for an applied ML/NLP challenge; rapid prototyping and model evaluation under time constraints.

Physics Olympics — 1st Place *University of British Columbia* Mar. 2022
Led a team of **8** IB students; placed **1st** among **40+** schools across BC.

PUBLICATIONS

Reimaging Health Equity, Equality, and Justice: A Fresh Vision Longwoods Insights (**2024**).
McCallum, E; Scala, N; Mason, T; ***Zheng, H****; Priore, A; Coppinger, T; Sochaniwskyj, M; Pettit A; Hill, MA; Shearkhani, S.*
longwoods.com/content/27405

Opportunities and Challenges for the Application of AI in Evaluations of Healthcare Services Longwoods Insights (**2024**).
Mateus, L; Qu, Y; ***Zheng, H****; Shearkhani, S*
longwoods.com/content/27544